MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

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INTRODUCTION

Mixing calculations involve computing the ratios in which two or more end-members are mixed in a sample. Mixing calculations are useful for a number of tasks in hydrology, such as hydrograph separation, water or solute mass balances, and identification of groundwater recharge sources. Most methods available for computing mixing ratios are based on assuming that end-member concentrations are perfectly known, which is rarely the case. Often end-members cannot be sampled and their concentrations vary in time and space. Still, much information about them is contained in the mixtures. To take advantage of this information, we present ***MIX,*** a maximum likelihood method to estimate mixing ratios, while acknowledging uncertainty in end-member concentrations. Maximizing the likelihood of concentration measurements with respect to both mixing ratios and end-member concentrations leads to a general constrained optimization problem. Results allow us to conclude that the method outperforms traditional approaches, such as least squares or linear mixing, in the computation of mixing ratios. ***MIX*** also yields improved estimates of end-members concentrations, thus enlarging the potential of mixing calculations. ***MIX*** requires defining the reliability of measurements, but results are quite robust with respect to the assumed standard deviations. A nice feature of ***MIX*** is that it allows us improving the quality of computations by increasing the number of samples and/or analyzed species.

For more information about methodology:
CARRERA, J., E. VÁZQUEZ-SUÑÉ, O. CASTILLO, AND X. SÁNCHEZ-VILA (2004)
A METHODOLOGY TO COMPUTE MIXING RATIOS WITH UNCERTAIN END-MEMBERS.
WATER RESOURCES RESEARCH, 40, DOI:10.1029/2003WR002263.

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

Input files: One Input file is needed

Format: ASCII file
filename = root
file extension is "prn"

Ex: "**filename.prn**"

Output files: 2 Output files

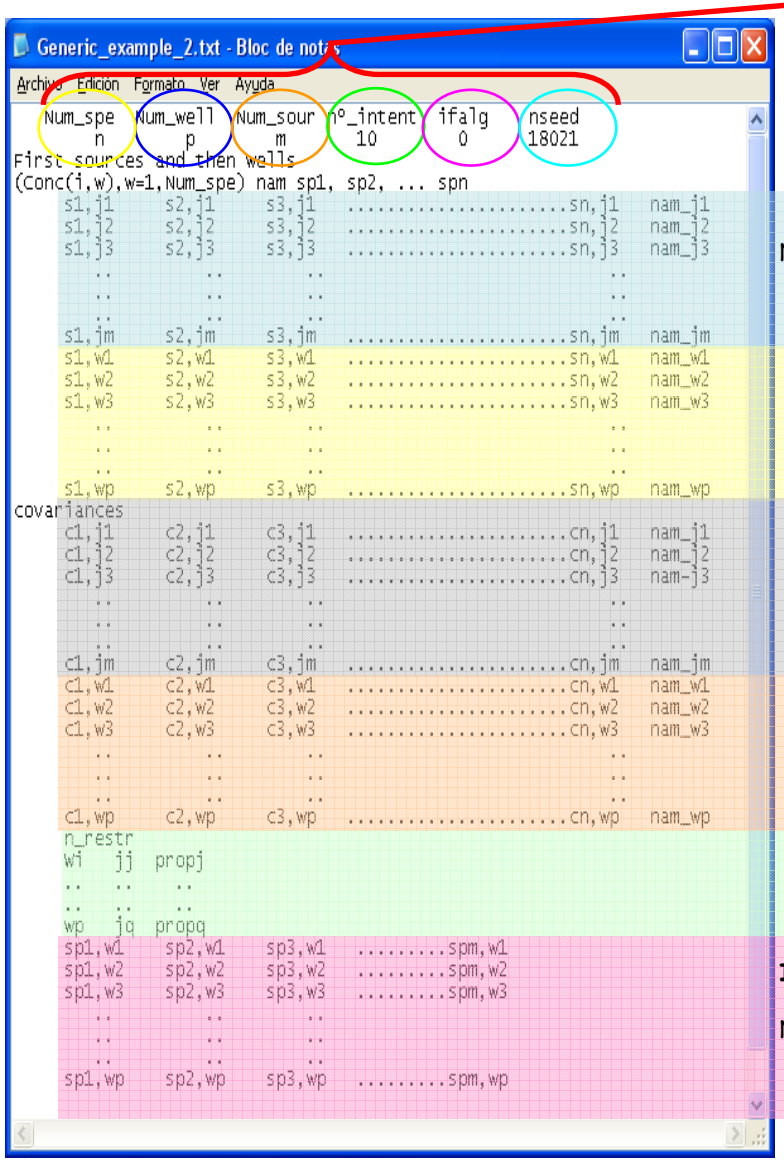
Format: ASCII file
filename = root
file extension is "res"

Ex: "**filename.res**"
"**filenameeigen.res**"

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

Grup d'hidrologia subterrània UPC CSIC

INPUT FILES GENERIC EXAMPLE



num_spe	num_well	num_sour	nº intent	flag	nseed
number of specie	number of wells	number of sources	number of attempts	0	any number

Concentrations

Num_spe (**n**) x Num_sour (**m**)	(s1,j1) ... (sn,jm) Concentration specie1 in source 1 ... Concentration specie n in source m
Num_spe (**n**) x Num_well (**p**)	(s1,w1) ... (sn,wp) Concentration specie1 in well 1 ... Concentration specie n in well p

Covariances

Num_spe (**n**) x Num_sour (**m**)	(c1,j1) ... (cn,jm) Covariance specie 1 in source1 ... Covariance specie n in source m
Num_spe (**n**) x Num_well (**p**)	(s1,w1) ... (sn,wp) Covariance specie1 in well 1 ... Covariance specie n in well p
Num_ constrains Well (i), Sour (j), prop (i)	(num_constrains) Number of source proportion restrictions (wi, jj, propj)....(wp,jq,propq) Well number i, Source number j, Source j proportion in the well i .... Well number p, Source number q, Source q proportion in the well p

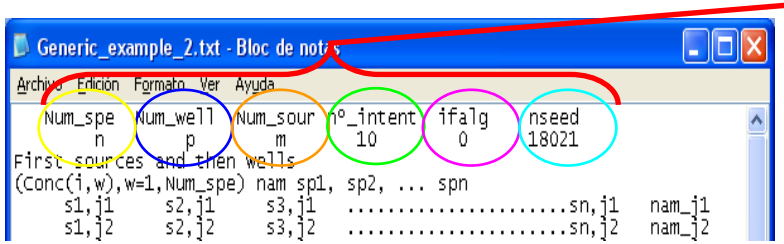
Input mixing ratios

Num_sour (**m**) x Num_well (**p**)	(sp1,w1) ... (spm,wp) Proportion source 1 in well 1 ... Proportion source m in well p [These values are a first approximated solution given by user]

MIX PROGRAM
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INPUT FILES GENERIC EXAMPLE



num_spe	num_well	num_sour	n° intent	flag	nseed
number of specie	number of wells	number of sources	number of attempts	0	any number

NUMBER OF ATTEMPTS:

1 = [Attempt from: input mixing ratios]

2 = [1]+[Attempt from: least squares mixing ratios]

3 = [1]+[2]+[Attempt from: same proportion for sources in wells]

4 = [1]+[2]+[3]+[Attempt from: input mixing ratios with random noise*]

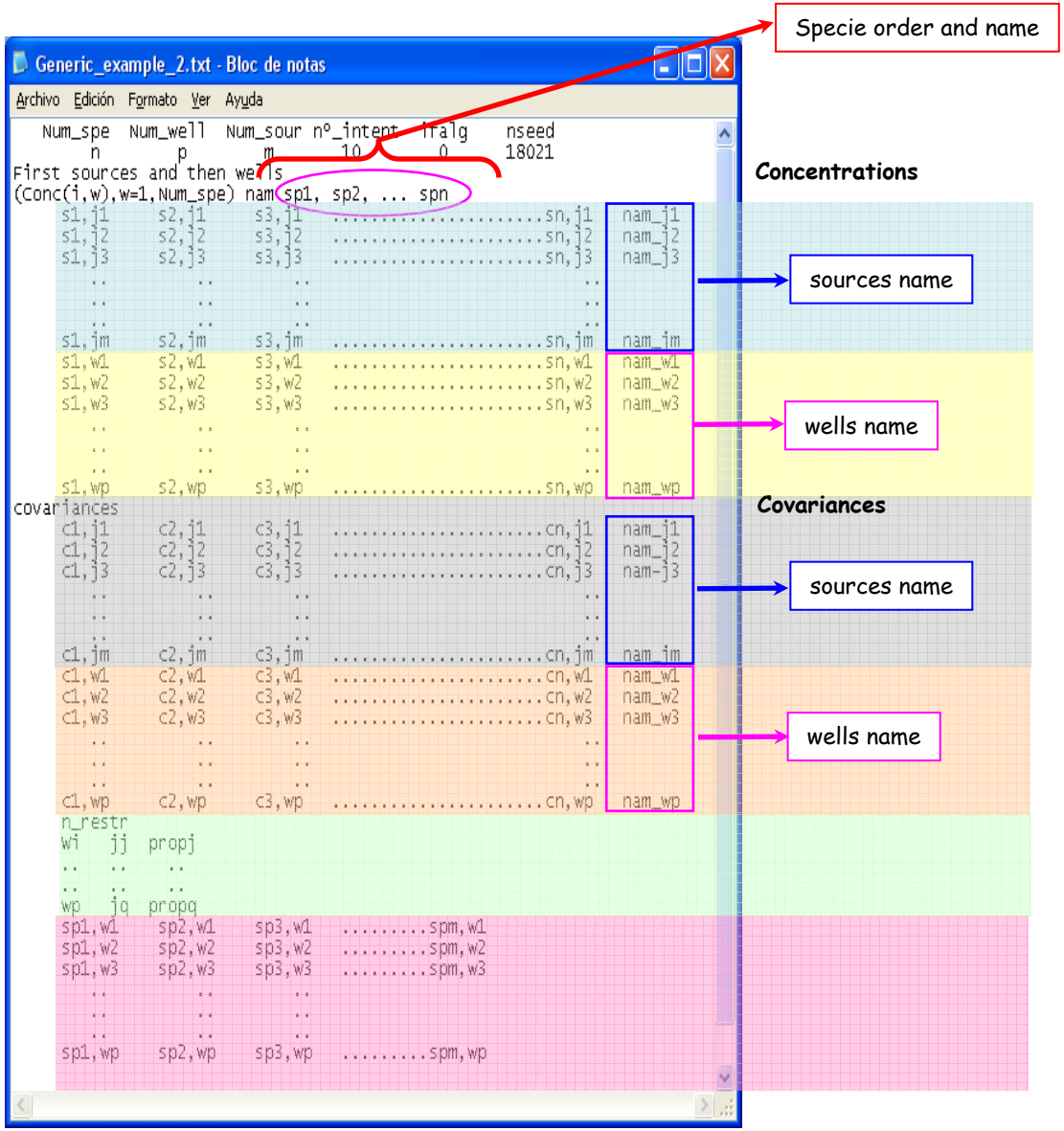
n = [1]+[2]+[3]+ ...+[n] [Attempt from: input mixing ratios with random noise*]

(*) Random noise is a random factor between +/- 0.25 that multiplies each input mixing ratio
Random noise initial value depends on "nseed".

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

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INPUT FILES GENERIC EXAMPLE



MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

INPUT FILES

Example 1:
(file: aa4.prn)

5 specie
4 wells
3 Sources
9 attempts
0 restrictions

aa4.prn - Bloc de notas

Archivo Edición Formato Ver Ayuda

```
  Num_spe  Num_well  Num_sour  nº_inten  iflag  nseed
        5         4         3         9      0  18201
 First sources and then wells
(CONC(i,j),j=1,Num_spe) nam sp1,sp2,sp3,sp4,sp5
      500.89450       807.33472       130.29805       788.71979       237.97029  Sour_1
       82.23440        66.22243       554.31506       274.93958       628.26062  Sour_2
      733.04761       394.85715       797.56995       664.48767        17.66558  Sour_3
      550.52484       411.22662       648.76434       563.66235       194.71411  Well_1
      301.49957       330.60428       362.23471       440.21863       463.07953  Well_2
      576.29169       402.26633       693.09833       562.80817       178.38208  Well_3
      429.62680       396.80719       453.40265       524.82141       323.36151  Well_4
 Covariances
 10000.00000000  10000.00000000   2500.00000000  10000.00000000   2500.00000000  Sour_1
  2500.00000000   2500.00000000  10000.00000000   2500.00000000  10000.00000000  Sour_2
 10000.00000000  10000.00000000  10000.00000000  10000.00000000    625.00000000  Sour_3
    16.00000000     16.00000000     16.00000000     16.00000000     16.00000000  Well_1
    16.00000000     16.00000000     16.00000000     16.00000000     16.00000000  Well_2
    16.00000000     16.00000000     16.00000000     16.00000000     16.00000000  Well_3
    16.00000000     16.00000000     16.00000000     16.00000000     16.00000000  Well_4
     0
        0.19134         0.26344         0.54522
        0.21803         0.58162         0.20035
        0.14196         0.19169         0.66635
        0.26401         0.47259         0.26340
```

Concentrations

Num_spe (5) x Num_sour (3)

Num_spe (5) x Num_well (4)

Covariances

Num_spe (5) x Num_sour (3)

Num_spe (5) x Num_well (4)

Num_ constrains = 0

Input mixing ratios
Num_sour (3) x Num_well (4)

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

INPUT FILES

Example 2: (file: aa4_b.prn)

5 specie
4 wells
3 Sources
9 attempts
2 restrictions

aa4_b.prn - Bloc de notas

Archivo Edición Formato Ver Ayuda

```
  Num_spe  Num_well  Num_sour  nº_inten  iflag  nseed
        5         4         3         9      0  18201
 First sources and then wells
(CONC(i,j),j=1,Num_spe) nam sp1,sp2,sp3,sp4,sp5
      500.89450       807.33472       130.29805       788.71979       237.97029       Sour_1
       82.23440        66.22243       554.31506       274.93958       628.26062       Sour_2
      733.04761       394.85715       797.56995       664.48767        17.66558       Sour_3
      550.52484       411.22662       648.76434       563.66235       194.71411       well_1
      301.49957       330.60428       362.23471       440.21863       463.07953       well_2
      576.29169       402.26633       693.09833       562.80817       178.38208       well_3
      429.62680       396.80719       453.40265       524.82141       323.36151       well_4
 Covariances
 10000.00000000  10000.00000000   2500.00000000  10000.00000000   2500.00000000       Sour_1
  2500.00000000   2500.00000000  10000.00000000   2500.00000000  10000.00000000       Sour_2
 10000.00000000  10000.00000000  10000.00000000  10000.00000000    625.00000000       Sour_3
    16.00000000     16.00000000     16.00000000     16.00000000     16.00000000       well_1
    16.00000000     16.00000000     16.00000000     16.00000000     16.00000000       well_2
    16.00000000     16.00000000     16.00000000     16.00000000     16.00000000       well_3
    16.00000000     16.00000000     16.00000000     16.00000000     16.00000000       well_4
     2
     1    2   0.0
     2    3   0.0
     0.20000         0.26000         0.54000
     0.22000         0.58000         0.20000
     0.14000         0.19000         0.67000
     0.26000         0.47000         0.27000
```

Concentrations

Num_spe (5) x Num_sour (3)

Num_spe (5) x Num_well (4)

Covariances

Num_spe (5) x Num_sour (3)

Num_spe (5) x Num_well (4)

Num_constrains = 2
well (1), source (2), prop=0.0
well (2), source (3), prop=0.0

Input mixing ratios

Num_sour (3) x Num_well (4)

MIX PROGRAM
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INPUT FILES

Example 3:
(file: eje10.prn)

5 specie
10 wells
3 Sources
9 attempts
0 restrictions

eje10.prn - Bloc de notas

Archivo Edición Formato Ver Ayuda

```
  Num_spe  Num_well  Num_sour  nº_inten  iflag  nseed
        5        10         3         9      0    345
 First sources and then wells
(CONC(i,j),j=1,Num_spe) nam sp1,sp2,sp3,sp4,sp5
      249.37444      139.87614      146.81291       15.41344       18.59980 Sour_1
      999.95660      193.81909       52.43275        5.06370       28.86399 Sour_2
      210.92389       89.75257      385.18024       28.71617       35.41355 Sour_3
      200.03552      200.15407      120.04971       19.92916       14.86655 well_1
     1000.03180      100.07006       89.93877        5.00816       10.03186 well_2
      306.09354      108.13609      281.87936       22.93295       29.01007 well_3
      493.14880      123.94719      192.01804       16.87007       20.42141 well_4
      270.95172      177.99934      145.10954       19.54019       16.93113 well_5
      207.94347       94.07504      340.95822       26.44659       34.39626 well_6
      831.04865      107.92040      124.00733        9.10785       13.47295 well_7
      483.98840      111.99564      219.95770       17.97619       22.96566 well_8
      430.99188      158.07959      138.99707       16.62249       15.99932 well_9
      581.98651      125.90985      161.12094       14.56948       17.39580 well_10
 Covariances
           1.00           1.00           1.00           1.00           1.00 Sour_1
           1.00           1.00           1.00           1.00           1.00 Sour_2
           1.00           1.00           1.00           1.00           1.00 Sour_3
     0.01000000     0.01000000     0.01000000     0.01000000     0.01000000 well_1
     0.01000000     0.01000000     0.01000000     0.01000000     0.01000000 well_2
     0.01000000     0.01000000     0.01000000     0.01000000     0.01000000 well_3
     0.01000000     0.01000000     0.01000000     0.01000000     0.01000000 well_4
     0.01000000     0.01000000     0.01000000     0.01000000     0.01000000 well_5
     0.01000000     0.01000000     0.01000000     0.01000000     0.01000000 well_6
     0.01000000     0.01000000     0.01000000     0.01000000     0.01000000 well_7
     0.01000000     0.01000000     0.01000000     0.01000000     0.01000000 well_8
     0.01000000     0.01000000     0.01000000     0.01000000     0.01000000 well_9
     0.01000000     0.01000000     0.01000000     0.01000000     0.01000000 well_10
              0
        0.99010        0.00000        0.00990
        0.00000        1.00000        0.00000
        0.23446        0.22830        0.53724
        0.24390        0.42632        0.32978
        0.80171        0.10243        0.09586
        0.12140        0.08516        0.79344
        0.12099        0.75866        0.12034
        0.16331        0.40228        0.43441
        0.60914        0.29976        0.09110
        0.31653        0.49892        0.18455
```

Concentrations

Num_spe (5) x Num_sour (3)

Num_spe (5) x Num_well (10)

Covariances

Num_spe (5) x Num_sour (3)

Num_spe (5) x Num_well (10)

Num_constrains = 0

Input mixing ratios

Num_sour (3) x Num_well (10)

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

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INPUT FILES

Example 4:
(file: eje10_b.prn)

5 specie
10 wells
3 Sources
9 attempts
4 restrictions

eje10_b.prn - Bloc de notas

Archivo Edición Formato Ver Ayuda

```
|   Num_spe   Num_well   Num_sour   nº_inten   iflag   nseed
        5         10          3          9        0      345
 First sources and then wells
(CONC(i,j),j=1,Num_spe) nam sp1,sp2,sp3,sp4,sp5
      249.37444       139.87614       146.81291       15.41344       18.59980       Sour_1
      999.95660       193.81909        52.43275        5.06370       28.86399       Sour_2
      210.92389        89.75257       385.18024       28.71617       35.41355       Sour_3
      200.03552       200.15407       120.04971       19.92916       14.86655       Well_1
     1000.03180       100.07006        89.93877        5.00816       10.03186       Well_2
      306.09354       108.13609       281.87936       22.93295       29.01007       Well_3
      493.14880       123.94719       192.01804       16.87007       20.42141       Well_4
      270.95172       177.99934       145.10954       19.54019       16.93113       Well_5
      207.94347        94.07504       340.95822       26.44659       34.39626       Well_6
      831.04865       107.92040       124.00733        9.10785       13.47295       Well_7
      483.98840       111.99564       219.95770       17.97619       22.96566       Well_8
      430.99188       158.07959       138.99707       16.62249       15.99932       Well_9
      581.98651       125.90985       161.12094       14.56948       17.39580       Well_10
 Covariances
           1.00            1.00            1.00            1.00           1.00       Sour_1
           1.00            1.00            1.00            1.00           1.00       Sour_2
           1.00            1.00            1.00            1.00           1.00       Sour_3
     0.01000000      0.01000000      0.01000000      0.01000000     0.01000000       Well_1
     0.01000000      0.01000000      0.01000000      0.01000000     0.01000000       Well_2
     0.01000000      0.01000000      0.01000000      0.01000000     0.01000000       Well_3
     0.01000000      0.01000000      0.01000000      0.01000000     0.01000000       Well_4
     0.01000000      0.01000000      0.01000000      0.01000000     0.01000000       Well_5
     0.01000000      0.01000000      0.01000000      0.01000000     0.01000000       Well_6
     0.01000000      0.01000000      0.01000000      0.01000000     0.01000000       Well_7
     0.01000000      0.01000000      0.01000000      0.01000000     0.01000000       Well_8
     0.01000000      0.01000000      0.01000000      0.01000000     0.01000000       Well_9
     0.01000000      0.01000000      0.01000000      0.01000000     0.01000000       Well_10
        4
        1         1       0.0
        4         3       0.0
        6         2       0.0
       10         1       0.0
      0.99010         0.00000         0.00990
      0.00000         1.00000         0.00000
      0.23446         0.22830         0.53724
      0.24390         0.42632         0.32978
      0.80171         0.10243         0.09586
      0.12140         0.08516         0.79344
      0.12099         0.75866         0.12034
      0.16331         0.40228         0.43441
      0.60914         0.29976         0.09110
      0.31653         0.49892         0.18455
```

Concentrations

Num_spe (5) x Num_sour (3)

Num_spe (5) x Num_well (10)

Covariances

Num_spe (5) x Num_sour (3)

Num_spe (5) x Num_well (10)

Num_constrains = 4
well (1), source (1), prop=0.0
well (4), source (3), prop=0.0
well (6), source (2), prop=0.0
well (10), source (1), prop=0.0

Input mixing ratios

Num_sour (3) x Num_well (10)

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

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INPUT FILES

Example 5:
(file: eje30.prn)

12 specie
7 wells
3 Sources
99 attempts
3 restrictions

eje30.prn - Bloc de notas

Archivo Edición Formato Ver Ayuda

```
  Num_spe  Num_well  Num_sour nº_intent     ifalg     nseed
       12         7         3        99         0      5555
First sources and then wells
(Conc(i,j),j=1,Num_spe) nam Cl,SO4,34S,F,O18,D,N,Br,EDTA,Zn,RA,B
    96.50    108.45      1.63    490.00     -5.91    -38.50      2.51      0.30      0.30    181.00     -2.58     48.00  REC
   380.00    208.60      3.15   1100.00     -5.95    -41.20     28.40      6.67    379.50    220.00     -2.08    683.33  RIU
 21050.00   2910.00     44.52   1000.00      1.50      8.20      0.84    120.00      0.00     10.00    -20.82   5000.00  MAR
   189.00    119.00      1.81     55.00     -6.70    -46.00      0.31      3.20      7.90     93.86     -0.02    150.00  PUIG
   211.86    220.72      1.81    197.00     -5.31    -40.04      5.83      0.54      7.90     19.50     -3.13    528.00  SCO1
   216.97    172.35      1.81    464.00     -5.41    -39.40      4.62      0.49      7.90     62.33     -2.58    634.00  REGA
   205.00    238.60      3.61    216.00     -6.20    -41.00     12.01      2.66      7.20     73.00     -2.57    233.00  COLA
   243.00    301.90      4.57    253.00     -6.80    -48.00     20.64      2.56      4.80     67.20     -2.17    264.00  COVA
   259.00    314.10      4.76    267.00     -7.10    -50.00     21.96      1.75      5.40     54.00     -2.73    149.00  FRIG
   696.00     10.00      0.15    228.00     -2.30    -24.00      0.42     16.80      1.50    185.00      1.40    430.00  TITN
covariances
 9.412E+01 1.186E+02 3.657E-02 9.704E+03 1.397E+00 5.930E+01 1.252E+00 1.008E+00 1.000E+02 3.048E+03 2.763E-01 3.074E+02  REC
 1.300E+04 3.917E+03 2.491E+00 3.026E+05 3.183E+00 1.528E+02 2.026E+02 5.004E+00 3.611E+04 3.108E+04 4.351E+00 1.168E+05  RIU
 1.000E+00 1.000E+00 1.000E-02 1.000E+02 1.000E-04 1.000E-02 1.000E+00 1.000E+00 1.000E+02 1.000E+02 1.000E-02 1.000E+02  MAR
 1.430E+03 1.416E+04 1.315E+01 4.850E+04 1.796E+00 2.117E+01 5.486E+02 1.922E+00 1.156E+02 2.302E+03 2.260E+00 2.260E+04  PUIG
 4.498E+02 4.882E+02 3.295E+00 9.802E+03 2.538E+00 1.443E+02 5.443E+02 1.026E+00 1.624E+02 1.951E+02 2.465E+00 1.125E+04  SCO1
 1.884E+03 2.980E+02 3.295E+00 2.253E+03 2.634E+00 1.397E+02 3.428E+02 1.022E+00 1.624E+02 1.071E+03 1.670E+00 1.618E+04  REGA
 4.213E+02 5.703E+02 1.403E-01 4.299E+03 3.845E-01 1.682E+01 3.705E+01 1.637E+00 1.130E+02 1.432E+03 1.666E+00 5.439E+04  COLA
 5.915E+02 9.124E+02 2.185E-01 7.401E+02 4.625E-01 2.305E+01 1.075E+02 1.590E+00 1.058E+02 1.229E+03 1.186E+00 6.980E+04  COVA
 6.038E+03 9.876E+02 2.365E-01 8.129E+02 5.042E-01 2.501E+01 1.216E+02 1.276E+00 1.073E+02 8.290E+02 1.871E+00 2.230E+04  FRIG
 4.844E+07 1.000E+04 2.260E+00 5.199E+06 5.290E+02 5.760E+04 1.904E+01 2.823E+04 3.250E+02 3.423E+06 1.960E+02 1.849E+07  TITN
         3
         1         3       0.0
         2         3       0.0
         3         3       0.0
     0.000     0.333     0.666  PUIG
     0.500     0.200     0.300  SCO1
     0.100     0.100     0.800  REGA
     0.450     0.540     0.010  COLA
     0.200     0.770     0.030  COVA
     0.700     0.290     0.010  FRIG
     0.450     0.500     0.050  TITN
```

Concentrations

Num_spe (12) x Num_sour (3)

Num_spe (12) x Num_well (7)

Covariances

Num_spe (12) x Num_sour (3)

Num_spe (12) x Num_well (7)

Num_constrains = 3
well (1), source (3), prop=0.0
well (2), source (3), prop=0.0
well (3), source (3), prop=0.0

Input mixing ratios

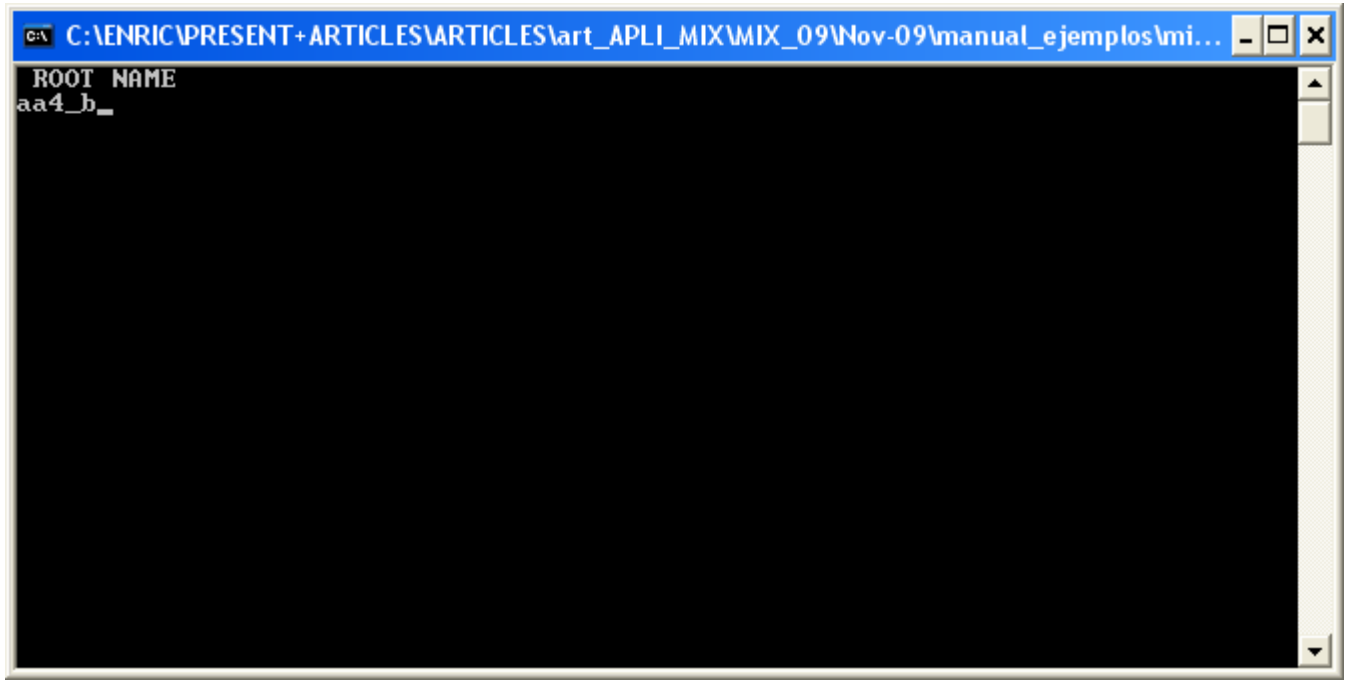
Num_sour (3) x Num_well (7)

MIX PROGRAM
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RUN (Input file and Mix_Program must be in the same directory):

1.- DOUBLE CLICK ON MIX_PROGRAM ICON. Appears a MSDOS screen.

2.- WRITE THE "ROOT" FILENAME



3.- RETURN. When computation is finished MSDOS screen is closed and a output file is written in the same directory as Input file and Mix_Program.

OUTPUT FILES

OUTPUT FILE STRUCTURE (*.RES)(1):

DATA INPUT VERIFICATION

Problem dimensions
Concentration data
Variances of concentration data
Constrains
Input mixing ratios
EigenAnalysis
Constrained least squares mixing ratios (initial)

ITERATIONS OUTPUT

Attempts

OPTIMAL SOLUTION

Best objective function
Computed values of sources concentrations
Computed mixing ratios
Objective function contributions

(1) If you have a previous calculation (*.res) with the same name, the new run will rewrite it.

MIX PROGRAM
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OUTPUT FILES Example 2 (file: aa4_b.res)

DATA INPUT VERIFICATION

aa4_b.res - Bloc de notas

Archivo Edición Formato Ver Ayuda

```
 DIMENSIONS
  Number of species:  5
  Number of wells:  4
  Number of sources:  3
  IOPTINI:  9
    iflag:  0
     seed:     18201

 CONCENTRATION DATA
  Species      sp1          sp2          sp3          sp4          sp5
  Sources (end-members):
  ----------------------
Sour_1       500.894      807.335      130.298      788.720      237.970
Sour_2       82.2344      66.2224      554.315      274.940      628.261
Sour_3       733.048      394.857      797.570      664.488      17.6656
  Wells (mixed water samples):
  ----------------------------:
Well_1       550.525      411.227      648.764      563.662      194.714
Well_2       301.500      330.604      362.235      440.219      463.080
Well_3       576.292      402.266      693.098      562.808      178.382
Well_4       429.627      396.807      453.403      524.821      323.362

 VARIANCES OF CONCENTRATION DATA
  Species      sp1          sp2          sp3          sp4          sp5
  Sources (end-members):
  ----------------------
Sour_1       10000.0      10000.0      2500.00      10000.0      2500.00
Sour_2       2500.00      2500.00      10000.0      2500.00      10000.0
Sour_3       10000.0      10000.0      10000.0      10000.0      625.000
  Wells (mixed water samples):
  ----------------------------:
Well_1       16.0000      16.0000      16.0000      16.0000      16.0000
Well_2       16.0000      16.0000      16.0000      16.0000      16.0000
Well_3       16.0000      16.0000      16.0000      16.0000      16.0000
Well_4       16.0000      16.0000      16.0000      16.0000      16.0000

 Num. fixed constraints:  2
 Mixing ratio in well   1 (Well_1    ),source 2 (Sour_2    ) fixed at  0.000000
 Mixing ratio in well   2 (Well_2    ),source 3 (Sour_3    ) fixed at  0.000000

 INPUT MIXING RATIOS
  Sources      Sour_1       Sour_2       Sour_3
  INPUT DATA WARNING: at well  1 sum of PROPs equals 0.74000     I correct it
Well_1      0.270270     0.00000      0.729730
  INPUT DATA WARNING: at well  2 sum of PROPs equals 0.80000     I correct it
Well_2      0.275000     0.725000     0.00000
Well_3      0.140000     0.190000     0.670000
Well_4      0.260000     0.470000     0.270000

 EIGENANALYSIS
 -------------
(see eigen file for details)

 CONSTRAINED LEAST SQUARES MIXING RATIOS (INITIAL)
 Sources          1            2            3
  Well:  1 0.238524     0.00000      0.761476
  Well:  2 0.392671     0.607329     0.00000
  Well:  3 0.103980     0.210965     0.685054
  Well:  4 0.339065     0.387435     0.273500

 ***************************************************************************
```

PROBLEM DIMENSIONS

CONCENTRATION DATA

COVARIANCES OF CONCENTRATION DATA

CONSTRAINS

INPUT MIXING RATIOS

EIGENANALYSIS

LEAST SQUARE MIXING RATIOS

1/8 (NEXT)

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

OUTPUT FILES Example 2 (file: aa4_b.res) **ITERATIONS PROCESS: ATTEMP 1**

aa4_b.res - Bloc de notas

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```
**************************************************************************
Attempt number:   1
**************************************************************************

INITIAL MIXING RATIOS
Sources          1             2             3
 Well:  1 0.270270      0.00000      0.729730
 Well:  2 0.275000      0.725000      0.00000
 Well:  3 0.140000      0.190000      0.670000
 Well:  4 0.260000      0.470000      0.270000

ITERATION PROCESS
-------------------------------------------------------------------------
 iter  line      alfa      objective       gradient    directional
  num serch                 function           norm     derivative
----- ----- --------- -------------- -------------- --------------
    1     0 0.100E+01  -207.0626       6182589.
    2     0 0.100E+01  -180.8286       5470014.       20.18940
    2     1 0.235E+01  -161.1491       4526914.       10.03655
    2     2 0.284E+01  -156.8370       4117573.       7.652786

   ..    ..    ..          ..             ..             ..
   ..    ..    ..          ..             ..             ..

   12     6 0.711E+00  -19.73343       305708.2       4.070098
   12     7 0.213E+01  -23.21567       1037093.       5.459397
   12     8 0.640E+01  -74.98962       8026713.       1.367586
   12     9 0.192E+02  -245.3182       6532475.       28.37308
   12    -1 0.790E-01  -19.38260       127195.0       3.359025
   13     0 0.100E+01  -20.12479       417242.6       4.539193
   13     0 0.200E-01  -19.38328       134952.2       2.301598
   13     0 0.400E-03  -19.38261       130866.1       2.299542
 Stop because of small change in obj. func.

  STOP ITERATIONS

objective function:   -19.3826
Expected value of sources concentrations
   409.91020   507.40265   206.50953   591.41640   340.87948
   143.95248    76.23585   583.92176   227.93496   636.31232
   690.07278   323.44604  1052.24106   541.10558    57.43235

Mixing ratios
    0.479      0.000      0.521
    0.588      0.412      0.000
    0.424      0.000      0.576
    0.621      0.157      0.222

**************************************************************************
Attempt number:   2
```

First Initial solution (INPUT MIXING RATIOS)

Iterative process

Objective function → differences between:

initial solution respect input mixing ratios
and
Computed concentration respect input concentrations

Final objective function

Computed values of sources concentrations

Computed mixing ratios

2/8 (NEXT)

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

OUTPUT FILES Example 2 (file: aa4_b.res)

ITERATIONS PROCESS: ATTEMP 2

aa4_b.res - Bloc de notas

Archivo Edición Formato Ver Ayuda

```
**************************************************************************
Attempt number:   2
**************************************************************************

INITIAL MIXING RATIOS
Sources          1           2           3
 Well:  1 0.238524     0.00000     0.761476
 Well:  2 0.392671     0.607329     0.00000
 Well:  3 0.103980     0.210965     0.685054
 Well:  4 0.339065     0.387435     0.273500

ITERATION PROCESS
-------------------------------------------------------------------------
 iter  line      alfa      objective       gradient    directional
  num serch              function           norm     derivative
----- ----- --------- ------------- ------------- -------------
    1     0 0.100E+01  -215.0210       8717019.
    2     0 0.100E+01  -188.9230       7787909.       23.76420
    2     1 0.544E+01  -135.6214       322562.3      -10.69796
    3     0 0.100E+01  -129.2340       250677.3       2.640433

   ..    ..    ..          ..             ..             ..
   ..    ..    ..          ..             ..             ..

    9    -1 0.160E+00  -19.52044       193281.8       4.449033
   10     0 0.100E+01  -20.14601       413293.4       5.421588
   10     0 0.200E-01  -19.52063       201452.7       3.113074
   10     0 0.400E-03  -19.52044       198266.6       3.118021
 Stop because of small change in obj. func.

  STOP ITERATIONS

objective function:   -19.5204
Expected value of sources concentrations
  413.19075   524.51758   181.08945   603.51473   341.37883
  152.58533    75.39729   597.53156   230.20978   620.57528
  677.44908   317.33923  1040.40770   533.93516    66.78568

Mixing ratios
    0.458     0.000     0.542
    0.566     0.434     0.000
    0.403     0.000     0.597
    0.593     0.172     0.235

**************************************************************************
```

Second Initial solution (INPUT MIXING RATIOS)

Iterative process

Objective function → differences between:

initial solution respect input mixing ratios
and
Computed concentration respect input concentrations

Final objective function

Computed values of sources concentrations

Computed mixing ratios

3/8 (NEXT)

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

OUTPUT FILES Example 2 (file: aa4_b.res) **ITERATIONS PROCESS: ATTEMP 3**

aa4_b.res - Bloc de notas

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```
**************************************************************************
Attempt number:   3
**************************************************************************

INITIAL MIXING RATIOS
Sources          1           2           3
 Well:   1 0.500000    0.00000     0.500000
 Well:   2 0.500000    0.500000     0.00000
 Well:   3 0.333333    0.333333    0.333333
 Well:   4 0.333333    0.333333    0.333333

ITERATION PROCESS
-------------------------------------------------------------------------
 iter  line      alfa      objective       gradient    directional
  num serch                 function           norm     derivative
----- ----- --------- -------------- -------------- --------------
    1     0 0.100E+01  -1711.304      0.3789142E+09
    2     0 0.100E+01  -1598.383      0.3091746E+09   94.68494
    2     1 0.304E+01  -1475.214      0.2387148E+09  -13.62521
    3     0 0.100E+01  -1339.008      0.2084069E+09   119.5395

   ..    ..    ..          ..             ..             ..
   ..    ..    ..          ..             ..             ..

   26     0 0.200E-01  -83.87448       63152.94       1.013084
   26     0 0.400E-03  -83.87298       63127.15       1.012815
   26     0 0.100E+01  -83.87380       50663.78       4.176988
   26     0 0.200E-01  -83.87295       63130.41      0.7873652
 Stop because of small change in obj. func.

  STOP ITERATIONS

objective function:   -83.8729
Expected value of sources concentrations
  301.49231   331.24560   360.52352   440.84647   461.25123
  163.13371   339.07767    90.25803   419.29555   597.09720
  722.38242   453.67272   858.14159   637.86392    20.71674

Mixing ratios
    0.407     0.000     0.593
    1.000     0.000     0.000
    0.347     0.000     0.653
    0.000     0.525     0.475

**************************************************************************
```

Second Initial solution (INPUT MIXING RATIOS)

Iterative process

Objective function → differences between:

initial solution respect input mixing ratios
and
Computed concentration respect input concentrations

Final objective function

Computed values of sources concentrations

Computed mixing ratios

4/8 (NEXT)

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

OUTPUT FILES Example 2 (file: aa4_b.res)

ITERATIONS PROCESS: ATTEMP n

aa4_b.res - Bloc de notas

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```
********************************************************************************
Attempt number:   n
********************************************************************************

INITIAL MIXING RATIOS
Sources          1             2             3
 Well:   1 0.460799      0.00000      0.539201
 Well:   2 0.580816      0.419184      0.00000
 Well:   3 0.490585      0.00000      0.509415
 Well:   4 0.515124      0.176599      0.308278

ITERATION PROCESS
-------------------------------------------------------------------------
 iter  line      alfa      objective      gradient    directional
  num serch                 function          norm     derivative
----- ----- --------- -------------- -------------- --------------
    1     0 0.100E+01  -340.1170      0.2487001E+08
    2     0 0.100E+01  -300.2062      0.2005642E+08   31.56422
    2     1 0.362E+01  -228.3171      0.1793455E+08   13.23996
    2     2 0.544E+01  -276.7663       3810738.       61.47141

   ..    ..    ..          ..             ..             ..
   ..    ..    ..          ..             ..             ..

   42     0 0.200E-01  -83.86957       58829.15       1.045155
   42     0 0.400E-03  -83.86815       58755.17       1.045401
   42     0 0.100E+01  -83.87441       48824.58       4.238642
   42     0 0.200E-01  -83.86812       58804.42      0.8099172
 Stop because of small change in obj. func.

  STOP ITERATIONS

objective function:   -83.8681
Expected value of sources concentrations
  301.48522   331.23426   360.51657   440.83665   461.29342
  161.53231   338.83304    87.73498   418.76907   598.81691
  720.51110   453.15685   855.92076   637.00426    22.58953

Mixing ratios
    0.404     0.000     0.596
    1.000     0.000     0.000
    0.344     0.000     0.656
    0.000     0.522     0.478

********************************************************************************
```

n Initial solution: Input mixing ratios with random noise

Iterative process

Objective function ➔differences between:

initial solution respect input mixing ratios
and
Computed concentration respect input concentrations

Final objective function

Computed values of sources concentrations

Computed mixing ratios

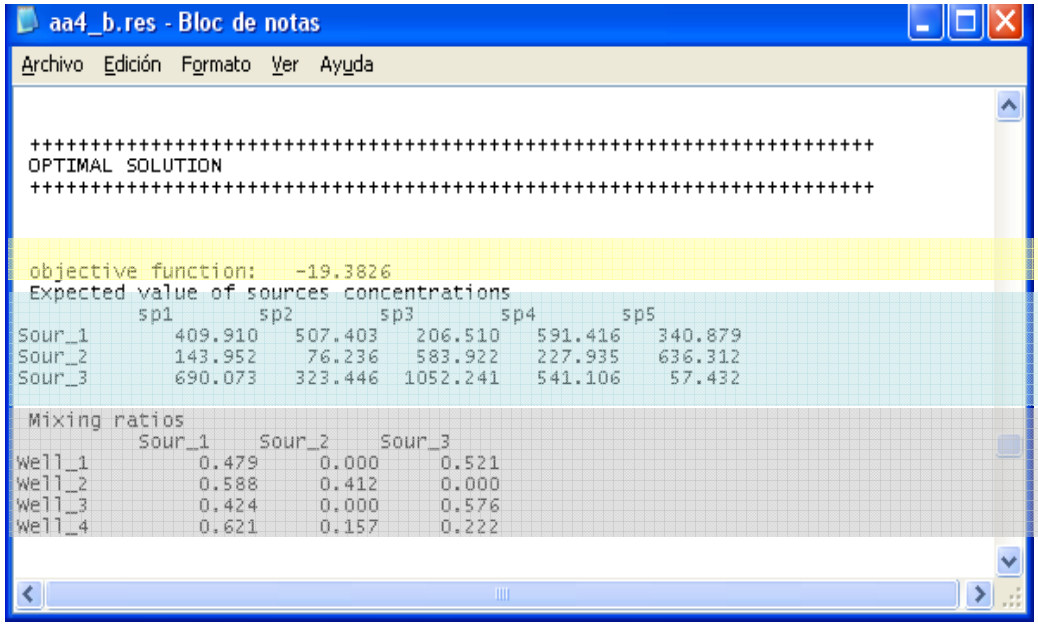
5/8 (NEXT)

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

OUTPUT FILES Example 2 (file: aa4_b.res)

OPTIMAL SOLUTION (I)

Comparing iterations results the program brings the optimal solution



BEST OBJECTIVE FUNCTION

COMPUTED VALUES OF SOURCES CONCENTRATIONS

COMPUTED MIXING RATIOS

6/8 (NEXT)

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

OUTPUT FILES Example 2 (file: aa4_b.res)

OPTIMAL SOLUTION (II)

OBJECTIVE FUNCTION CONTRIBUTION

aa4_b.res - Bloc de notas

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```
 Species:sp1              Contrib. to obj.fcnt:  -2.9775          (%)   15.362
 ------------------
     Water Conc.Estim  Conc.Measrd  Difference NormlzdDiff
   ------- ----------- -----------  ---------- -----------
Sour_1          409.910     500.894     -90.984      -0.910
Sour_2          143.952      82.234      61.718       1.234
Sour_3          690.073     733.048     -42.975      -0.430
Well_1          555.857     550.525       5.332       1.333
Well_2          300.301     301.500      -1.199      -0.300
Well_3          571.348     576.292      -4.943      -1.236
Well_4          430.256     429.627       0.630       0.157

    ::    ::    ::          ::           ::           ::
    ::    ::    ::          ::           ::           ::
    ::    ::    ::          ::           ::           ::
    ::    ::    ::          ::           ::           ::
    ::    ::    ::          ::           ::           ::
    ::    ::    ::          ::           ::           ::

 Species:spn              Contrib. to obj.fcnt:  -3.5202          (%)   18.162
 ------------------
     Water Conc.Estim  Conc.Measrd  Difference NormlzdDiff
   ------- ----------- -----------  ---------- -----------
Sour_1          340.879     237.970     102.909       2.058
Sour_2          636.312     628.261       8.052       0.081
Sour_3           57.432      17.666      39.767       1.591
Well_1          193.222     194.714      -1.492      -0.373
Well_2          462.636     463.080      -0.443      -0.111
Well_3          177.549     178.382      -0.833      -0.208
Well_4          324.441     323.362       1.079       0.270

  Contributions to objective function
  -----------------------------------

          sp1       sp2       sp3       sp4       sp5          total
  Contributions of sources
Sour_1   -0.414    -4.498    -1.162    -1.946    -2.118   -10.138
Sour_2   -0.762    -0.020    -0.044    -0.442    -0.003    -1.271
Sour_3   -0.092    -0.255    -3.243    -0.761    -1.265    -5.616
  Contributions of wells
Well_1   -0.888    -0.004    -0.089    -0.075    -0.070    -1.125
Well_2   -0.045    -0.025    -0.001    -0.061    -0.006    -0.138
Well_3   -0.764    -0.023    -0.017    -0.005    -0.022    -0.831
Well_4   -0.012    -0.118    -0.001    -0.095    -0.036    -0.263
```

SPECIE 1 OBJECTIVE FUNCTION CONTRIBUTION

Water concentration estimated
Water concentration measured
Difference between estimated and measured concentrations
Normalized difference between estimated and measured

SPECIE n OBJECTIVE FUNCTION CONTRIBUTION

Water concentration estimated
Water concentration measured
Difference between estimated and measured concentrations
Normalized difference between estimated and measured

SOURCES AND WELLS OBJECTIVE FUNCTION CONTRIBUTION BY SPECIE AND TOTAL

7/8 (NEXT)

MIX PROGRAM
INPUT AND OUTPUT FILES DESCRIPTION

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OUTPUT FILES Example 2 (file: aa4_beigen.res)

OPTIMAL SOLUTION (II)

EIGENANALYSIS

aa4_beigen_MANUAL.prn - Bloc de notas

Archivo Edición Formato Ver Ayuda

```
*** WARNING  ERROR 1 from DEPISF.  The performance index =
***          1.8691190452233335D+13.  A value greater than 100 is considered
***          poor.

Eigen analysis of raw sources data
----------------------------------

              EVAL
      1           2           3           4
-410.4      -64.3        10.3         0.0

                 EVEC
          1          2          3          4
1    0.6590   -0.3575   -0.6616    0.0166
2    0.4841   -0.4719    0.7368   -0.0131
3    0.5757    0.8059    0.1380   -0.0007
4   -0.0042    0.0004    0.0207    0.9998

Performance index = ******

sample no.:  1  Well_1
--------------

                  EVAL
          1           2           3           4
-187863.8    -22737.2         1.0         0.0

               EVEC
         1        2        3        4
1    0.684    0.730    0.000    0.000
2    0.000    0.000    1.000    0.000
3    0.730   -0.684    0.000    0.000
4    0.000    0.000    0.000    1.000

Performance index =  0.011
 Sour_1      Sour_2      Sour_3

poor performance: eigenvalue= 0.10000E+01 of combination:
  0.000       1.000       0.8955E-16

sample no.:  2  Well_2
--------------
  .........     ..........
  .........     ..........
  .........     ..........

sample no.:  p  Well_p
--------------

                  EVAL
          1           2           3           4
-207523.1    -33262.3    -19110.4         0.0

               EVEC
         1        2        3        4
1    0.636   -0.568   -0.522    0.000
2    0.333    0.812   -0.479    0.000
3    0.696    0.131    0.706    0.000
4    0.000    0.000    0.000    1.000

Performance index =  0.025
```

Well 1 EigenValues

Well 1 EigenVectors

Well p EigenValues

Well p EigenVectors